

Jannat

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Education

Jan. 2026 – present	Ph.D., Computer Science Virginia Commonwealth University, Richmond, VA
Aug. 2024 – Dec. 2025	M.S., Computer Science Virginia Commonwealth University, Richmond, VA Thesis: <i>Relationship Extraction Using Retrieval Augmented Generation for Biomedical Datasets</i>
Jul. 2023 – May 2024	Master of Computer Applications CHRIST University, Bangalore, India
Jul. 2019 – May 2022	B.S., Applied Physical Science University of Delhi, New Delhi, India

Honors & Awards

Summer 2025	cMEDA Summer Student Scholarship (\$3,000). Center for Microbiome Engineering & Data Analysis (cMEDA), Virginia Commonwealth University - awarded to support a summer research project with Dr. Bridget T. McInnes.
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Research Interests

- Machine Learning, Multimodal AI, and Natural Language Processing
- Vision–language tools for understanding scientific charts and figures (structured information extraction)
 - Robustness of multimodal models to noisy, real-world inputs
 - Accessibility tools for visual data, aimed at blind and low-vision users
 - Retrieval-augmented generation and biomedical text mining

Research Experience

Jan. 2026 – present	Graduate Research Assistant (Ph.D.). Bioinformatics & Machine Learning Lab, Virginia Commonwealth University. Building a vision–language tool using models such as ViT (Vision Transformer) and DePlot to make scientific figures and visual data accessible to blind and low-vision users.
Jan. 2025 – Dec. 2025	Research Assistant. Natural Language Processing Lab, Virginia Commonwealth University. Developed Retrieval-Augmented Generation (RAG) pipelines with LLMs for relationship extraction from 1,500+ biomedical abstracts; work culminated in an MS thesis and the BioASQ 2025 shared-task paper below.

Publications

1. **Jannat**, Dil C, Arodz T and McInnes BT (2026) Biomedical retrieval-augmented generation for relation classification. *Frontiers in Research Metrics and Analytics* 11:1919972. doi: [10.3389/frma.2026.1919972](https://doi.org/10.3389/frma.2026.1919972)
2. Taylor S, Dil C, Shah A, **Jannat**, Oldham C, Upadhyay A, Varughese J, Yazbeck N and McInnes BT (2025) [NLP@VCU at BioASQ 2025: information extraction on the GutBrainIE dataset](#). In: *Working Notes of CLEF 2025 – Conference and Labs of the Evaluation Forum*. CEUR Workshop Proceedings, Vol. 4038.
3. **Jannat** (2025) [Relationship extraction using retrieval augmented generation for biomedical datasets](#). M.S. Thesis. Richmond, VA: Virginia Commonwealth University.

Projects

- Adversarial Training of YOLO for Aerial Imagery** [\[code\]](#) : Python, PyTorch, YOLOv11
- Tennis Video Analysis System** [\[code\]](#) : Python, YOLOv8, OpenCV, Streamlit

RideShare, Real-Time Carpooling App [\[code\]](#) : React Native, Firebase, Node.js, OpenStreet Map

Teaching Experience

Sep. 2025 – Dec. 2025	Grading Assistant. <i>Advanced Algorithms</i>
Sep. 2025 – Dec. 2025	Teaching Assistant. <i>Introduction to Engineering</i>
Sep. 2024 – Dec. 2024	Teaching Assistant. <i>Programming Using C</i>
Sep. 2024 – Dec. 2024	Teaching Assistant. <i>Introduction to Engineering</i>

Professional Experience

Jun. 2025 – Jul. 2025	Graduate Student Software Engineer. Engineering Career Services, Virginia Commonwealth University. Built and deployed a student–alumni mentoring interface (Google Apps Script) used by 200+ students.
Oct. 2021 – Feb. 2022	Software Engineer Intern. Raceme Tenders LLP, Delhi, India. Migrated a legacy PHP/MySQL site to AWS in a three-person team; built a web crawler extracting 500+ documents daily.

Leadership & Volunteering

Feb. 2025 – Apr. 2025	Data Analyst (Volunteer), Women of Connections (nonprofit organization).
Jul. 2019 – May 2021	Core Member, Computer Science Club. University of Delhi
Jul. 2021 – May 2022	Vice President, Computer Science Club. University of Delhi
Jul. 2020 – May 2022	Volunteer, Enactus Society. University of Delhi

References

Dr. Tomasz Arodz, Associate Professor of Computer Science, Virginia Commonwealth University
Dr. Bridget T. McInnes, Associate Professor of Computer Science, Virginia Commonwealth University
Trishabh Chadda, Principal Engineering Manager, Microsoft